OS & SP

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SKILL – 15

1.Skill15.c

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GNU nano 8.7.1 skill15.c *

```c
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/wait.h>

int main(void)
{
    int fd;
    pid_t pid;

    /* Open combined.txt for writing.
       O_TRUNC clears old contents. */
    fd = open(
        "combined.txt",
        O_WRONLY | O_CREAT | O_TRUNC,
        0644
    );

    if (fd == -1) {
        perror("Error opening combined.txt");
        return 1;
    }

    /* Create child process */
    pid = fork();

    if (pid < 0) {
        perror("fork");
        close(fd);
        return 1;
    }

    /* Child process */
    if (pid == 0) {
        /* Redirect stdout to combined.txt */
        if (dup2(fd, STDOUT_FILENO) == -1) {
            perror("dup2 stdout");
            _exit(1);
        }

        /* Redirect stderr to the same file */
        if (dup2(fd, STDERR_FILENO) == -1) {
            perror("dup2 stderr");
            _exit(1);
        }

        /* Close original descriptor */
        close(fd);

        /* Write to stdout */
        dprintf(
            STDOUT_FILENO,
            "This is standard output.\n"
        );

        /* Write to stderr */
        dprintf(
```

```c
        /* Write to stdout */
        dprintf(
            STDOUT_FILENO,
            "This is standard output.\n"
        );

        /* Write to stderr */
        dprintf(
            STDERR_FILENO,
            "This is standard error.\n"
        );

        /* Execute a command producing both outputs */
        execlp(
            "sh",
            "sh",
            "-c",
            "echo Command output; echo Command error >&2",
            (char *)NULL
        );
        /* Runs only if execlp fails */
        perror("execlp");
        _exit(1);
    }
    /* Parent process */
    close(fd);
    /* Wait for child */
    waitpid(pid, NULL, 0);
    printf("Q15 completed successfully.\n");
    printf("stdout and stderr were merged into combined.txt\n");
    return 0;
}
```

OUTPUT :

```
lohitha27@LAPTOP-A9F545FG:~/SKILL$ nano skill15.c
lohitha27@LAPTOP-A9F545FG:~/SKILL$ gcc -Wall -Wextra -g skill15.c -o skill15
lohitha27@LAPTOP-A9F545FG:~/SKILL$ ./skill15
Q15 completed successfully.
stdout and stderr were merged into combined.txt
lohitha27@LAPTOP-A9F545FG:~/SKILL$
```